

F24-107-D-SparkQuest

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Chapter 1

Introduction

1.1 Problem Statement

As education is evolving rapidly, traditional learning methods are now falling short. Most of the time these traditional learning methods fail to engage a young learner's mind, due to which a student loses focus. A primary-grade student requires an interactive and dynamic learning environment to enhance their learning [2]. Such a platform not only captures a student's interest but also fulfills the need for diverse learning styles. There are multiple products in the market that can be used to engage a student but they mostly target a single aspect of learning and do not fully engage a student. Moreover, there is a gap in existing educational tools that offer audio-visual aids and an interactive learning environment for the student. The gap is quite notable when it comes to regions like Pakistan, where not everyone can speak English. Such language barriers affect the learning of students as they are unable to grasp the concepts fully. A tool that caters to the students' bilingual needs and also provides them with an interactive learning environment by combining different aspects of interactive learning is crucial for the students, especially in the Pakistan region. An application like this can bridge the gaps, enhance creativity, and engage students in critical thinking[3].

1.2 Scope

The scope of SparkQuest lies in developing a user-friendly dynamic educational website that integrates tools like Generative AI and RAG. SparkQuest targets the students of grades one to five and it focuses on three major subjects English, Maths, and Science. A core functionality of this app is the multi-modal question-answering module where students can ask questions about a document, image, or video. Additionally, this feature allows students to upload the content they want to ask questions from. Moreover, the web

app also provides the student with text-to-audio support, aiding auditory learners. Bilingual support for Urdu and English language is also incorporated in the web app. Another crucial feature is the inclusion of a parental/guardian role where the parent/guardian can monitor their child's progress.

With SparkQuest, we hope to make education fun for young leaders. The project focuses on making use of AI in the field of education to make it more fun for primary-grade students. With the advancements in the field of AI, we can make education a way of fostering creativity among students where they can learn in a collaborative environment.

1.3 Modules

SparkQuest, an E-learning web app, is based on a total of seven modules. These modules are divided into two modes, User mode and Evaluation mode. The modules are interlinked components at the back end that will provide a complete learning experience to a primary-grade student to promote additional self-learning.

1.3.1 Module 1 - Two modes of the Client Web Application

This module will consist of two modes, User mode and Evaluation mode. The user mode will be for guided learning of students where they can learn from the material by asking questions or getting summaries from it. The evaluation mode will be a test-based module for the students with add-on features like Parental controls to ensure that the student learning curve is supervised.

1. Student Mode
2. Evaluation Mode
3. Parental Controls (Refer to Module 4)

1.3.2 Module 2 - Question Answering(QA) in User Mode

In the user QA mode, students will make use of various types of media to ask questions and understand ambiguous concepts, helping them understand and describe what is happening in the content. These different types of media include text, PDF files, images, and videos. Here, a user can either upload these media files or choose from an existing database provided in the web application.

1. Text-based Question Answering via RAG

2. Visual Question Answering from images and videos via RAG

1.3.3 Module 3 - Question Answering from User in Evaluation Mode

Followed by the question-answering in the user mode (refer to Module 2), the student can enter the evaluation mode through the UI where they will be accessed through a series of questions about their understanding from generated content (questions answered) of that topic. This will help them analyze their understanding of a topic and aid in preparing for tests.

1. Question generation and its types (Short answers, Fill in the blanks, MCQs, etc)
2. Evaluation of answers
3. Timed Module

1.3.4 Module 4 - Parental/Guardian Role in Evaluation Mode

This module introduces a dedicated interface for parents or guardians to monitor the student's progress and interactions with the chatbot. The module allows them to view all conversations between the student and the chatbot across Module 2 and Module 3. By reviewing these chats, parents or guardians can help the student correct any misunderstandings or inaccuracies. This feature acts as an additional safeguard, ensuring that the student is learning accurate information and promoting a collaborative learning environment between the child and their guardians.

1. Report generation of complete Chat

1.3.5 Module 5 - Audio Input and Output

As primary-grade students may not be very fluent in English typing, therefore this module will allow them to provide audio input and receive an audio output. For example, a student can ask the web app through audio support to produce a summary or a question. The web app will process this audio input and generate text-based results, followed by a prompt that takes user preference of how they want the results, i.e. text only or text along with audio. Users can set this in the default options of the application also.

1. Audio Input Processing
2. Audio Output Processing

3. Default settings of Audio Input/Output
4. Integration with all other Modules

1.3.6 Module 6 - Dual language Support

The geographic audience of the E-learning website is limited to the region of Pakistan, hence the dual language support will incorporate Urdu language. This module will be integrated with Module 5, where a user will be provided with the option to provide an Urdu audio input and receive audio output in Urdu.

1. Urdu Speech to Text
2. Urdu Text to Speech
3. Urdu to English context understanding
4. Urdu English Translation

1.3.7 Module 7 - Database

An extensive database will be maintained for this website. This will record data of the user and parent/guardian including their Grade. Moreover, all questions answers, and explanations will also be recorded here. This personalized data of students will be used to generate questions (refer to module 3) and improve the quality of responses over time. The schema will cover the following aspects:

1. Student details
2. Parent/Guardian details
3. Questions generated
4. Questions asked by the user and their corresponding answers generated

1.4 User Classes and Characteristics

The user classes and characteristics involved in SparkQuest are primary-grade students, parents/guardians, and developers. The table 1.1 explains how every user is associated with our system and how they are going to use our application.

Table 1.1: User cases involved in SparkQuest and their descriptions

User class	Description
Primary Grade Students	A Primary Grade Student is a student between grades 1 to 5 who interacts with SparkQuest to enhance their learning abilities and outcomes. There are more than 20 million students across Pakistan in this age group [1]. A Primary Grade student will use features such as question answering and audio-visual aids specifically for the subjects English, Maths, and Science. Using both guided learning and self-paced evaluations, these students will get to experience a spectrum of possibilities in education.
Parents/Guardians	A Parent or Guardian is an adult responsible for overseeing a student's learning and interaction with the web application. These users will use the guardian feature to oversee their child's progress and monitor their activity.
Developers	A developer is responsible for updating and developing the SparkQuest platform. They will build the back-end system ensuring a seamless integration of Artificial Intelligence(AI) and Retrieval Augmented Reality (RAG). Moreover, they are expected to work on the accuracy of the web application.

Chapter 2

Project Requirements

This chapter outlines the functional and non-functional requirements for the SparkQuest system by detailing the various features that are necessary for the system to accomplish the intended functionalities and general performance of the platform. The functional requirements revolve around the major functionalities such as the question-answering module, Evaluation Mode, Content Management, and Parental Monitoring to explain how the system engages the students, how the inputs are processed, and how the educational contents and feedback are delivered. Further, Non-functional requirements provide details about the system's performance, ease of use, security, extensibility, and dependability necessary to give a comfortable environment to students and parents. To this end, this chapter will present a current working specification of the system to give the reader a holistic view of its features, as well as the technical and end-user requirements that must be met for the project to be workable.

2.0.1 Usecase Diagram

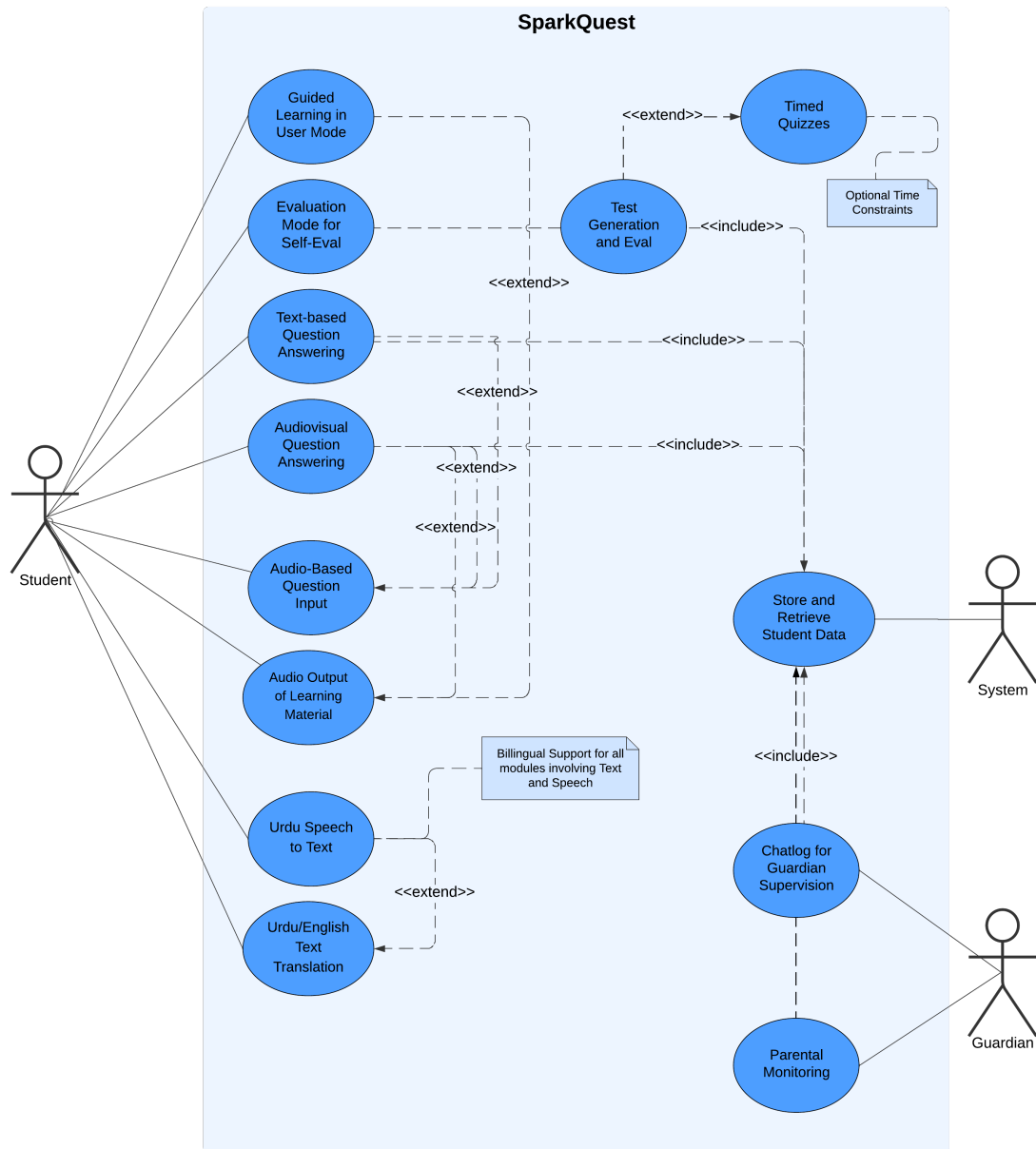


Figure 2.1: The use case diagram of SparkQuest depicting the interaction of Student and Guardian with the system

SparkQuest caters to diverse user interactions through different modules and features. Each feature is mapped to a specific use case as shown in figure 2.1. The use-case diagram can help us identify the level of interaction students and parents/guardians will have with the system.

2.0.2 High Level Use Cases

1. UC-1

Table 2.1: UC-1: Guided Learning in User Mode

Attribute	Description
Use Case ID	UC-1
Use Case Name	Guided Learning in User Mode
Actors	Student
Description	A student accesses the User Mode to engage with uploaded or pre-existing educational content such as text, videos, or images. The system generates summaries and visualizations to assist in learning.
Preconditions	The student is logged into the system and has access to User Mode.
Postconditions	The student receives interactive summaries or visual explanations of the content.

2. UC-2

Table 2.2: UC-2: Evaluation Mode for Self-Assessment

Attribute	Description
Use Case ID	UC-2
Use Case Name	Evaluation Mode for Self-Assessment
Actors	Student
Description	The student selects a subject and takes quizzes or tests. The system evaluates the answers and provides feedback based on the student's performance.
Preconditions	The student must have access to Evaluation Mode.
Postconditions	The student receives detailed feedback, including scores and analysis of their answers.

3. UC-3

Table 2.3: UC-3: Parental Monitoring of Student Progress

Attribute	Description
Use Case ID	UC-3
Use Case Name	Parental Monitoring of Student Progress
Actors	Parent/Guardian
Description	Parents or guardians monitor the student's learning progress, including activity logs, test results, and learning patterns.
Preconditions	The parent/guardian must be logged into the parental portal.
Postconditions	The parent/guardian successfully reviews the student's activity, test results, and other performance indicators.

4. UC-4

Table 2.4: UC-4: Text-based Question Answering via RAG

Attribute	Description
Use Case ID	UC-4
Use Case Name	Text-based Question Answering via RAG
Actors	Student
Description	The student asks a text-based question related to their study materials, and the system generates answers using RAG (Retrieval-Augmented Generation).
Preconditions	The student has uploaded or selected relevant content.
Postconditions	The student receives a contextual answer based on the content.

5. UC-5

Table 2.5: UC-5: Visual Question Answering from Images and Videos via RAG

Attribute	Description
Use Case ID	UC-5
Use Case Name	Visual Question Answering from Images and Videos via RAG
Actors	Student
Description	The student uploads or selects images/videos and asks a question. The system analyzes the media and provides an answer based on its content.
Preconditions	The student has uploaded or selected relevant media content (image/video).
Postconditions	The student receives a visual or textual explanation based on the media content.

6. UC-6

Table 2.6: UC-6: Test Generation and Evaluation

Attribute	Description
Use Case ID	UC-6
Use Case Name	Test Generation and Evaluation
Actors	Student
Description	The student enters Evaluation Mode, where the system generates a test based on selected subject and difficulty level, evaluates the answers, and provides feedback.
Preconditions	The student must have access to Evaluation Mode and select the criteria for the test.
Postconditions	The student receives test results and feedback.

7. UC-7

Table 2.7: UC-7: Monitor Student Chat with Chatbot

Attribute	Description
Use Case ID	UC-7
Use Case Name	Monitor Student Chat with Chatbot
Actors	Parent/Guardian
Description	Parents or guardians view the conversations between the student and the chatbot, ensuring content appropriateness and engagement.
Preconditions	The parent must be logged into their account.
Postconditions	The parent successfully reviews all chatbot interactions.

8. UC-8

Table 2.8: UC-8: Audio-Based Question Input

Attribute	Description
Use Case ID	UC-8
Use Case Name	Audio-Based Question Input
Actors	Student
Description	The student asks questions using audio input, and the system processes the voice input, converts it to text, and then provides an answer.
Preconditions	Audio input features must be enabled for the student.
Postconditions	The student receives an answer based on the audio input.

9. UC-9

Table 2.9: UC-9: Audio-Based Output of Learning Material

Attribute	Description
Use Case ID	UC-9
Use Case Name	Audio-Based Output of Learning Material
Actors	Student
Description	The system presents learning materials, summaries, or answers through audio output, allowing the student to listen to the content.
Preconditions	Audio output feature is enabled.
Postconditions	The student receives the learning material through an audio format.

10. UC-10

Table 2.10: UC-10: Urdu Speech to Text

Attribute	Description
Use Case ID	UC-10
Use Case Name	Urdu Speech to Text
Actors	Student
Description	The student speaks in Urdu, and the system converts the speech to text, allowing for further interaction and processing.
Preconditions	The student provides Urdu audio input.
Postconditions	The system transcribes the Urdu audio into text for further use in question answering or evaluations.

11. UC-11

Table 2.11: UC-11: Text Translation between Urdu and English

Attribute	Description
Use Case ID	UC-11
Use Case Name	Text Translation between Urdu and English
Actors	Student
Description	The system translates text between Urdu and English to aid student understanding.
Preconditions	Text are provided in either Urdu or English.
Postconditions	The translated text is presented to the student.

12. UC-12

Table 2.12: UC-12: Store and Retrieve Student Learning Progress

Attribute	Description
Use Case ID	UC-12
Use Case Name	Store and Retrieve Student Learning Progress
Actors	System
Description	The system logs student progress, interactions, test results, and learning patterns for future retrieval by the student, teacher, or parent.
Preconditions	The student interacts with the learning platform.
Postconditions	Student progress are stored and can be retrieved when needed for further learning sessions or reports.

2.0.3 Extended Use Cases

1. EUC-1

Table 2.13: EUC-1: Guided Learning in User Mode

Attribute	Description
Use Case ID	EUC-1
Use Case Name	Guided Learning in User Mode
Actors	Student (Primary)
Description	The student accesses User Mode to upload or select pre-existing materials (text, images, or videos) and receives system-generated answers.
Trigger	The student initiates this use case by selecting or uploading content.
Preconditions	• The student is logged into the system. • The student has access to User Mode. • The student has relevant content to upload or select.
Postconditions	• The student receives answers to the material. • The system logs the interaction for parental monitoring.
Normal Flow	• The student logs into the system. • The student selects or uploads content. • The system processes the content using RAG. • The system provides answers based on the uploaded content. • The system logs the session for future review.
Alternate Flow	If the uploaded content is unsupported, the system displays an error and prompts the student to upload a valid file format.
Exceptions	If the system fails to process the content, it notifies the student and logs the error for future resolution.
Frequency of Use	This feature is used frequently during study sessions, depending on the student's learning needs.
Special Requirements	• The system must support various file formats (text, video, images).
Assumptions	The student has an internet connection.

2. EUC-2

Table 2.14: EUC-2: Evaluation Mode for Self-Assessment

Attribute	Description
Use Case ID	EUC-2
Use Case Name	Evaluation Mode for Self-Assessment
Actors	Student (Primary)
Description	The student enters Evaluation Mode to take quizzes based on their studied content, receiving feedback on their performance.
Trigger	The student selects Evaluation Mode from the dashboard.
Preconditions	• The student is logged into the system. • The student has access to Evaluation Mode.
Postconditions	• The student receives quiz feedback. • The system logs quiz results for progress tracking.
Normal Flow	• The student selects Evaluation Mode and chooses a subject. • The student selects a quiz type and difficulty level. • The system generates a quiz based on the selected options. • The student completes and submits the quiz. • The system evaluates the quiz and provides feedback. • The system logs the quiz results for future progress monitoring.
Alternate Flow	If the quiz is timed, the system shows a countdown timer during the quiz.
Exceptions	If quiz generation fails, the system notifies the student and logs the error.
Frequency of Use	This use case is accessed after studying content, typically weekly.
Special Requirements	• The system must support various question types (MCQs, short answers, fill-in-the-blanks). • Bilingual support should be available for quiz content in both English and Urdu.
Assumptions	The student has relevant content to evaluate their knowledge.

3. EUC-3

Table 2.15: EUC-3: Parental Monitoring of Student Progress

Attribute	Description
Use Case ID	EUC-3
Use Case Name	Parental Monitoring of Student Progress
Actors	Parent/Guardian (Primary)
Description	Parents monitor the student's activity logs, quiz results, and chatbot interactions to track progress.
Trigger	The parent logs into the parental control interface.
Preconditions	• The parent is logged into the system. • The student has interacted with the platform.
Postconditions	• The parent successfully reviews the student's progress. • The system logs the parental interaction for monitoring purposes.
Normal Flow	• The parent logs into the system. • The system retrieves the student's activity logs, quiz results, and chatbot logs. • The parent reviews the student's activity and progress. • The parent can download reports for future reference.
Alternate Flow	The parent filter logs by period (e.g., last week, last month).
Exceptions	If no logs are available, the system informs the parent and suggests checking back later.
Frequency of Use	Parents are expected to check progress weekly or monthly.
Special Requirements	• The system should provide secure access to sensitive student data. • The parental dashboard should offer bilingual support (English and Urdu).
Assumptions	The student has interacted with the system, generating progress logs.

4. EUC-4

Table 2.16: EUC-4: Text-based Question Answering via RAG

Attribute	Description
Use Case ID	EUC-4
Use Case Name	Text-based Question Answering via RAG
Actors	Student (Primary)
Description	The student asks questions based on study content, and the system provides answers using RAG.
Trigger	The student types a question after reviewing or uploading content.
Preconditions	• The student has uploaded or selected study content. • The student is logged into the system.
Postconditions	• The student receives an answer. • The interaction is logged for future reference.
Normal Flow	• The student types a question based on the content. • The system processes the question using RAG. • The system provides the answer to the student. • The interaction is logged for further monitoring.
Alternate Flow	If the system cannot find relevant content, it suggests alternative content.
Exceptions	If the question is out of scope, the system suggests rephrasing or uploading new content.
Frequency of Use	Students are expected to use this feature during study sessions.
Special Requirements	• The system should process questions in both English and Urdu. • Responses should be generated within a few seconds.
Assumptions	The student has study content uploaded or selected for reference.

5. EUC-5

Table 2.17: EUC-5: Visual Question Answering from Images and Videos via RAG

Attribute	Description
Use Case ID	EUC-5
Use Case Name	Visual Question Answering from Images and Videos via RAG
Actors	Student (Primary)
Description	The student asks questions about images or videos, and the system generates answers based on the media.
Trigger	The student uploads or selects a video or image and asks a question.
Preconditions	• The student has uploaded or selected media content. • The student is logged into the system.
Postconditions	• The student receives an answer. • The interaction is logged for future review.
Normal Flow	• The student uploads/selects an image or video. • The student asks a question about the media. • The system analyzes the media and generates an answer. • The interaction is logged for future reference.
Alternate Flow	If the media is too complex, the system requests clearer input.
Exceptions	If the media is unsupported, the system notifies the student to upload a supported file.
Frequency of Use	This feature is used when students interact with multimedia content.
Special Requirements	• The system should process multimedia files efficiently. • Bilingual support must be available for media-based responses.
Assumptions	The student has relevant media to upload and analyze.

6. EUC-6

Table 2.18: EUC-6: Test Generation and Evaluation

Attribute	Description
Use Case ID	EUC-6
Use Case Name	Test Generation and Evaluation
Actors	Student (Primary)
Description	The system generates quizzes based on the student's topic selections and evaluates their answers.
Trigger	The student selects Evaluation Mode and generates a quiz.
Preconditions	• The student is logged into Evaluation Mode. • The student has selected topics and test options.
Postconditions	• The student receives feedback and quiz results. • The system logs the results for progress tracking.
Normal Flow	• The student selects the subject and difficulty level. • The system generates a quiz. • The student completes and submits the quiz. • The system evaluates the quiz and provides feedback. • The quiz results are logged for future reference.
Alternate Flow	If the quiz is timed, the system auto-submits upon timeout.
Exceptions	If the system fails to generate a quiz, it prompts the student to retry later.
Frequency of Use	This feature is used periodically for self-assessment.
Special Requirements	• The system should generate dynamic quizzes with varying question types. • Bilingual support for both questions and answers.
Assumptions	The student has selected relevant topics for the quiz.

7. EUC-7

Table 2.19: EUC-7: Monitor Student Chat with Chatbot

Attribute	Description
Use Case ID	EUC-7
Use Case Name	Monitor Student Chat with Chatbot
Actors	Parent/Guardian (Primary)
Description	The parent view the conversation logs between the student and chatbot to monitor progress.
Trigger	The parent accesses the chat logs from the parental portal.
Preconditions	• The parent is logged into the system. • The student has interacted with the chatbot.
Postconditions	• The parent successfully reviews the chat logs.
Normal Flow	• The parent logs into the system. • The system retrieves the student's chat logs with the chatbot. • The parent reviews the logs and flags inappropriate content. • The session ends, and the logs are saved for future review.
Alternate Flow	The parent filters the logs by date or topic.
Exceptions	If no logs are available, the system informs the parent to check later.
Frequency of Use	Parents are expected to review chat logs weekly or monthly.
Special Requirements	• The system must securely store and display chat logs. • Bilingual chat logs should be available for both English and Urdu.
Assumptions	The student has engaged with the chatbot frequently enough to generate logs.

8. EUC-8

Table 2.20: EUC-8: Audio-Based Question Input

Attribute	Description
Use Case ID	EUC-8
Use Case Name	Audio-Based Question Input
Actors	Student (Primary)
Description	The student asks questions using audio input, and the system processes the voice input, converts it to text, and provides an answer.
Trigger	The student activates the audio input feature and speaks a question.
Preconditions	• The student is logged into the system. • Audio input feature must be enabled for the student.
Postconditions	• The student receives an answer based on the audio input. • The system logs the interaction for future reference.
Normal Flow	• The student asks a question using voice. • The system converts the audio to text. • The system processes the transcribed question using RAG. • The system provides an answer in text or audio form. • The student reviews the answer. • The system logs the interaction for progress tracking.
Alternate Flow	If the audio input is unclear, the system prompts the student to try again.
Exceptions	If the system fails to process the audio input, it notifies the student.
Frequency of Use	It will be used during study by students who prefer voice interaction.
Special Requirements	• The system must support bilingual audio input (English and Urdu). • Real-time audio-to-text conversion is required for smooth interaction.
Assumptions	• The student has access to a device capable of audio input.

9. EUC-9

Table 2.21: EUC-9: Audio-Based Output of Learning Material

Attribute	Description
Use Case ID	EUC-9
Use Case Name	Audio-Based Output of Learning Material
Actors	Student (Primary)
Description	The system presents learning materials, or answers through audio output, allowing the student to listen to the content.
Trigger	The student requests learning material via the audio output feature.
Preconditions	The student is logged in and the Audio output feature is enabled.
Postconditions	The student receives learning material in an audio format.
Normal Flow	- The student selects the content (learning material, summary, or answer) to be delivered via audio. - The system converts the text-based content to audio using text-to-speech capabilities. - The audio content is played for the student. - The student listens to the audio content. - The system logs the interaction for progress tracking.
Alternate Flow	If the student cannot hear the audio properly, the system allows for text-based alternatives.
Exceptions	If the system fails to generate audio content, it notifies the student and provides text-based content instead.
Frequency of Use	This feature will be used frequently by students who prefer listening to the material instead of reading.
Special Requirements	- The system must support bilingual audio output. - Generate real-time audio for a seamless learning experience.
Assumptions	- The student has access to a device capable of audio output. - The system can efficiently convert text to audio.

10. EUC-10

Table 2.22: EUC-10: Urdu Speech to Text

Attribute	Description
Use Case ID	EUC-10
Use Case Name	Urdu Speech to Text
Actors	Student (Primary)
Description	The student speaks in Urdu, and the system converts the speech to text for further processing or interactions.
Trigger	The student activates the Urdu audio input feature and speaks.
Preconditions	• The student is logged into the system. • The student provides Urdu audio input.
Postconditions	• The system transcribes the Urdu speech into text. • The system processes the transcribed text for further actions such as question answering or evaluations.
Normal Flow	• The student activates the Urdu audio input feature. • The student speaks in Urdu. • The system converts the speech to text. • The transcribed text is processed for further actions. • The interaction is logged for future reference.
Alternate Flow	If Urdu audio input is unclear, the system prompts the student to try again.
Exceptions	If the system cannot process the Urdu speech, it notifies the student and suggests trying text input instead.
Frequency of Use	Frequently by students who prefer interacting in Urdu.
Special Requirements	• The system must support Urdu speech-to-text conversion. • Real-time speech recognition is required for a smooth experience.
Assumptions	• The student has access to a microphone-enabled device. • The system is capable of processing Urdu audio inputs efficiently.

11. EUC-11

Table 2.23: EUC-11: Text Translation between Urdu and English

Attribute	Description
Use Case ID	EUC-11
Use Case Name	Text Translation between Urdu and English
Actors	Student (Primary)
Description	The system translates text between Urdu and English to assist the student in understanding the material.
Trigger	The student requests a text translation between Urdu and English.
Preconditions	The student has provided text in either Urdu or English.
Postconditions	- The translated text is presented to the student. - The translation is logged for progress tracking.
Normal Flow	- The student requests a Language translation. - The system provides the translated text to the student. - The student reviews the translated text. - The system logs the interaction for future reference.
Alternate Flow	If the translation is unclear, the system allows the student to refine the text input.
Exceptions	If the system cannot translate the text, it notifies the student and suggests using another language.
Frequency of Use	This feature will be used periodically when students need translations between the two languages.
Special Requirements	- The system must support both Urdu-to-English and English-to-Urdu translation. - Translations should be accurate and contextually appropriate.
Assumptions	- The student has provided the text to be translated. - The system has sufficient resources to process language translations efficiently.

12. EUC-12

Table 2.24: EUC-12: Store and Retrieve Student Learning Progress

Attribute	Description
Use Case ID	EUC-12
Use Case Name	Store and Retrieve Student Learning Progress
Actors	System
Description	The system logs student progress, interactions, test results, and learning patterns for future retrieval by the student, teacher, or parent.
Trigger	Student interacts with the system, triggering the logging of progress.
Preconditions	• The student is logged into the system. • The system has the storage capacity to log student progress.
Postconditions	• Student progress is logged and stored in the system. • The progress can be retrieved for parent monitoring.
Normal Flow	• The student engages with the platform (e.g., answering quizzes, asking questions). • The system logs the interaction and stores relevant data such as progress, scores, and session activity. • The system retrieves the data when required by the parent. • The system generates reports based on the stored progress.
Alternate Flow	Parent can access the progress logs at any time through an interface.
Exceptions	If there is an issue retrieving the data, the system notifies the user of the problem and suggests trying again later.
Frequency of Use	This feature runs continuously during all interactions and is accessed periodically for review.
Special Requirements	• The system must securely store student data and progress logs. • Data retrieval must be fast and reliable when generating reports.
Assumptions	• The student has interacted with the system and has logged data. • The system is capable of storing and retrieving data efficiently.

2.1 Functional Requirements

Functional requirements are the core features of a system. For SparkQuest, each requirement is organized into different modules. Each module focuses on providing a quality and interactive mode of education for the students. Moreover, the system also focuses on parental guidance so that parents can monitor their child's progress.

2.1.1 Module 1: Modes of the Web Application

1. FR1-001

Table 2.25: FR1-001: User Mode for Learning Assistance

Attribute	Description
Title	User Mode for Learning Assistance
ID	FR1-001
Version	1.0
Description	The system shall provide a User Mode that allows students to upload educational materials (text, PDF, images, videos) and receive learning assistance, including summaries and visualizations, based on the uploaded content.
Priority	High
Rationale	Ensures students can independently learn from uploaded materials and receive appropriate learning support.
Cross-Reference	Evaluation Mode (FR1-002), Parental Controls (FR1-003)

2. FR1-002

Table 2.26: FR1-002: Evaluation Mode for Quizzes and Tests

Attribute	Description
Title	Evaluation Mode for Quizzes and Tests
ID	FR1-002
Version	1.0
Description	The system shall provide an Evaluation Mode where students can take quizzes and tests based on their studied content. The tests will include short-answer questions and multiple-choice questions (MCQs).
Priority	High
Rationale	Provides students the opportunity to assess their understanding of studied material and prepare for exams.
Cross-Reference	User Mode (FR1-001), Parental Controls (FR1-003)

3. FR1-003

Table 2.27: FR1-003: Parental Controls for Monitoring Progress

Attribute	Description
Title	Parental Controls for Monitoring Progress
ID	FR1-003
Version	1.0
Description	The system shall allow for Parental Controls where a parent or guardian can monitor the student's progress, and receive activity logs.
Priority	Medium
Rationale	Enables parents/guardians to monitor their child's learning activities.
Cross-Reference	User Mode (FR1-001), Evaluation Mode (FR1-002)

2.1.2 Module 2: Question Answering (QA) in User Mode

1. FR2-001

Table 2.28: FR2-001: Text-Based Question Answering via RAG

Attribute	Description
Title	Text-Based Question Answering via RAG
ID	FR2-001
Version	1.0
Description	The system shall allow students to ask questions via text related to uploaded or selected content, and the system shall provide answers using Retrieval-Augmented Generation (RAG).
Priority	High
Rationale	Enables students to clarify doubts and understand ambiguous concepts by querying the system based on the content they are studying.
Cross-Reference	Visual Question Answering (FR2-002), User Mode (FR1-001)

2. FR2-002

Table 2.29: FR2-002: Visual Question Answering via RAG

Attribute	Description
Title	Visual Question Answering via RAG
ID	FR2-002
Version	1.0
Description	The system shall allow students to ask questions using visual content (videos and images), and the system shall provide answers based on the analysis of the uploaded media using Retrieval-Augmented Generation (RAG).
Priority	High
Rationale	Ensures students can query the system about images and videos for a better understanding of visual materials.
Cross-Reference	Text-Based Question Answering (FR2-001), User Mode (FR1-001)

2.1.3 Module 3: Question Answering in Evaluation Mode

1. FR3-001

Table 2.30: FR3-001: Interface for Evaluation Mode Quizzes

Attribute	Description
Title	Interface for Evaluation Mode Quizzes
ID	FR3-001
Version	1.0
Description	The system shall provide an interface for students to enter Evaluation Mode, where they can take quizzes with different question formats such as short answers, fill-in-the-blanks, and multiple-choice questions (MCQs).
Priority	High
Rationale	Provides a structured environment for students to test their knowledge and skills based on various types of questions.
Cross-Reference	Question Generation (FR3-002), Timed Assessments (FR3-004)

2. FR3-002

Table 2.31: FR3-002: Dynamic Question Generation by Subject and Difficulty

Attribute	Description
Title	Dynamic Question Generation by Subject and Difficulty
ID	FR3-002
Version	1.0
Description	The system shall generate questions dynamically, segmented by subject (English, Math, Science) and difficulty level (easy, medium, hard), based on the student's selected preferences.
Priority	High
Rationale	Ensures that students are tested on subject-specific material, with questions tailored to their skill level.
Cross-Reference	Evaluation Mode Interface (FR3-001), Answer Evaluation (FR3-003)

3. FR3-003

Table 2.32: FR3-003: Answer Evaluation and Instant Feedback

Attribute	Description
Title	Answer Evaluation and Instant Feedback
ID	FR3-003
Version	1.0
Description	The system shall evaluate student answers based on the correct answers and provide instant feedback.
Priority	High
Rationale	Ensures that students receive timely feedback on their performance, enabling them to identify areas of improvement.
Cross-Reference	Question Generation (FR3-002), Timed Assessments (FR3-004)

4. FR3-004

Table 2.33: FR3-004: Timed Assessments

Attribute	Description
Title	Timed Assessments
ID	FR3-004
Version	1.0
Description	The system shall allow for timed assessments, where students will be given a set time to complete the test.
Priority	Medium
Rationale	It Helps students manage their time effectively during tests and prepares them for real-world exam conditions.
Cross-Reference	Evaluation Mode Interface (FR3-001), Answer Evaluation (FR3-003)

2.1.4 Module 4: Parental/Guardian Role in Evaluation Mode

1. FR4-001

Table 2.34: FR4-001: Parental Monitoring Interface

Attribute	Description
Title	Parental Monitoring Interface
ID	FR4-001
Version	1.0
Description	The system shall provide parents/guardians with access to a monitoring interface where they can view student progress and interactions with the chatbot in User Mode and Evaluation Mode.
Priority	Medium
Rationale	Enables parents/guardians to monitor their child's learning activities and guide as needed.
Cross-Reference	Student Progress Report (FR4-002)

2. FR4-002

Table 2.35: FR4-002: Conversation Log Generation

Attribute	Description
Title	Conversation Log Generation
ID	FR4-002
Version	1.0
Description	The system shall generate and display complete logs of conversations between the student and the chatbot, covering interactions from Modules 2 and 3.
Priority	Medium
Rationale	Provides transparency for parents/guardians regarding their child's interactions with the system, ensuring accountability and oversight.
Cross-Reference	Parental Monitoring Interface (FR4-001)

2.1.5 Module 5: Audio Input and Output

1. FR5-001

Table 2.36: FR5-001: Audio Input Processing

Attribute	Description
Title	Audio Input Processing
ID	FR5-001
Version	1.0
Description	The system shall allow students to ask questions using audio input, which will be processed by the system to provide answers.
Priority	High
Rationale	Ensures accessibility for younger students who may not be proficient in typing or reading, allowing them to interact with the system via voice.
Cross-Reference	Audio Output (FR5-002), Audio Integration (FR5-004)

2. FR5-002

Table 2.37: FR5-002: Audio Output Processing

Attribute	Description
Title	Audio Output Processing
ID	FR5-002
Version	1.0
Description	The system shall allow students to receive audio output responses, providing audio-based answers to their questions.
Priority	High
Rationale	Improves accessibility by allowing students to receive spoken answers, especially beneficial for early learners.
Cross-Reference	Audio Input (FR5-001), Audio Integration (FR5-004)

3. FR5-003

Table 2.38: FR5-003: Audio Input/Output Settings Management

Attribute	Description
Title	Audio Input/Output Settings Management
ID	FR5-003
Version	1.0
Description	The system shall provide default settings for managing audio input and output preferences, enabling users to configure settings easily.
Priority	Medium
Rationale	Provides flexibility for students and parents to adjust the audio settings according to individual preferences or needs.
Cross-Reference	Audio Input (FR5-001), Audio Output (FR5-002)

4. FR5-004

Table 2.39: FR5-004: Integration of Audio Input/Output Across All Modules

Attribute	Description
Title	Integration of Audio Input/Output Across All Modules
ID	FR5-004
Version	1.0
Description	The system shall integrate the audio input and output functionality across all relevant modules, ensuring that both User Mode and Evaluation Mode support audio-based interactions.
Priority	High
Rationale	Ensures that audio features are available consistently throughout the application, improving the user experience for younger students.
Cross-Reference	Audio Input (FR5-001), Audio Output (FR5-002)

2.1.6 Module 6: Dual Language Support

1. FR6-001

Table 2.40: FR6-001: Urdu Speech-to-Text Conversion

Attribute	Description
Title	Urdu Speech-to-Text Conversion
ID	FR6-001
Version	1.0
Description	The system shall support Urdu speech-to-text conversion, allowing students to provide audio input in Urdu.
Priority	High
Rationale	Enables students who are more comfortable with Urdu to interact with the system using their native language through voice input.
Cross-Reference	Urdu Text-to-Speech (FR6-002), Translation Support (FR6-004)

2. FR6-002

Table 2.41: FR6-002: Urdu Text-to-Speech Functionality

Attribute	Description
Title	Urdu Text-to-Speech Functionality
ID	FR6-002
Version	1.0
Description	The system shall provide Urdu text-to-speech functionality, allowing students to receive audio responses in Urdu.
Priority	High
Rationale	Enhances accessibility by allowing students to receive audio responses in Urdu, making it easier for non-English speakers to understand.
Cross-Reference	Urdu Speech-to-Text (FR6-001), Translation Support (FR6-004)

3. FR6-003

Table 2.42: FR6-003: Urdu to English Context Understanding

Attribute	Description
Title	Urdu to English Context Understanding
ID	FR6-003
Version	1.0
Description	The system shall have context understanding capabilities for Urdu to English, ensuring that questions asked in Urdu can be processed in context to generate accurate answers.
Priority	High
Rationale	Ensures that language differences are accurately handled, allowing Urdu-speaking students to receive correct answers even when questions are asked in Urdu.
Cross-Reference	Translation Support (FR6-004)

4. FR6-004

Table 2.43: FR6-004: Urdu-English Translation Support

Attribute	Description
Title	Urdu-English Translation Support
ID	FR6-004
Version	1.0
Description	The system shall provide translations between Urdu and English, allowing users to toggle between languages during their interaction with the platform.
Priority	Medium
Rationale	Provides flexibility for users to switch between Urdu and English, promoting bilingual support for a broader user base.
Cross-Reference	Urdu Speech-to-Text (FR6-001), Urdu Text-to-Speech (FR6-002)

2.1.7 Module 7: Database

1. FR7-001

Table 2.44: FR7-001: Student Profile Storage

Attribute	Description
Title	Student Profile Storage
ID	FR7-001
Version	1.0
Description	The system shall store detailed student profiles in the database, including student ID, name, grade, and a reference to a Guardian's ID.
Priority	High
Rationale	Provides a centralized location for all student-related data, ensuring personalization of the learning experience.
Cross-Reference	Parent/Guardian Data Storage (FR7-002), Evaluation Reports (FR7-007)

2. FR7-002

Table 2.45: FR7-002: Parent/Guardian Data Storage

Attribute	Description
Title	Parent/Guardian Data Storage
ID	FR7-002
Version	1.0
Description	The system shall store parent/guardian details in the database, including unique identifiers, name, email/phone number.
Priority	Medium
Rationale	Ensures that parents/guardians are associated with the correct student, allowing access to their child's learning data and reports.
Cross-Reference	Student Profile Storage (FR7-001)

3. FR7-003

Table 2.46: FR7-003: Conversation Log Storage

Attribute	Description
Title	Conversation Log Storage
ID	FR7-003
Version	1.0
Description	The system shall store conversation logs between the student and the chatbot, ensuring that raw text for each student interaction is recorded.
Priority	Medium
Rationale	Provides a record of interactions for review by parents/guardians and future reference by the system.
Cross-Reference	Evaluation Logs (FR7-005), Evaluation Reports (FR7-007)

4. FR7-004

Table 2.47: FR7-004: Question and Answer Storage

Attribute	Description
Title	Question and Answer Storage
ID	FR7-004
Version	1.0
Description	The system shall store all questions asked by the user in the database, including question type (MCQs, short answers, fill in the blanks), and content along with corresponding student answers as raw text conversation.
Priority	High
Rationale	Ensures that the guardian can view all the student's progress.
Cross-Reference	Conversation Log Storage (FR7-003), Evaluation Reports (FR7-007)

5. FR7-005

Table 2.48: FR7-005: Uploaded Content Storage

Attribute	Description
Title	Uploaded Content Storage
ID	FR7-005
Version	1.0
Description	The system shall store all uploaded content, including text, images, videos, and PDFs, along with metadata such as content ID, subject (English, Math, Science), and source (uploaded by the student).
Priority	High
Rationale	Ensures that all educational materials are properly stored and categorized for easy retrieval and use in learning sessions.
Cross-Reference	Question and Answer Storage (FR7-004)

6. FR7-006

Table 2.49: FR7-006: Evaluation Report Generation and Storage

Attribute	Description
Title	Evaluation Report Generation and Storage
ID	FR7-006
Version	1.0
Description	The system shall generate and store evaluation reports in the database, ensuring that test results, scores, and feedback are retrievable by both the student and parent/guardian.
Priority	High
Rationale	Provides both students and parents with insights into performance, enabling review of test results and continuous learning improvements.
Cross-Reference	Question and Answer Storage (FR7-004)

2.2 Non-Functional Requirements

Non-functional requirements describe the overall quality of a system that ensures the system runs smoothly and reliably. SparkQuest's requirements include areas like usability, reliability, and accuracy. By fulfilling these requirements, we aim to enhance the quality of our system, making it a user-friendly platform for primary grade students.

2.2.1 Module 1: Modes of the Web Application

1. NFR1-001

Table 2.50: NFR1-001: Usability

Attribute	Description
Title	Usability
ID	NFR1-001
Version	1.0
Description	The system shall provide an intuitive and easy-to-navigate interface for both User Mode and Evaluation Mode, ensuring students can use the system with minimal assistance.
Metric	Usability tests with 90% of users aged 6-10 navigating with minimal guidance using React and JavaScript.
Priority	High
Rationale	Ensures young students can engage with the platform independently.
Originator	Development Team

2. NFR1-002

Table 2.51: NFR1-002: Performance

Attribute	Description
Title	Performance
ID	NFR1-002
Version	1.0
Description	The system shall load content (text, images, videos, PDFs) within 3-4 seconds on average.
Metric	Load testing will ensure content loads within 4 seconds for 95% of requests.
Priority	High
Rationale	SQLite for lightweight database operations ensures achievable performance targets without excessive server load.
Originator	Development Team

3. NFR1-003

Table 2.52: NFR1-003: Reliability

Attribute	Description
Title	Reliability
ID	NFR1-003
Version	1.0
Description	The system shall ensure 99% up-time, with REST API requests handled via FASTAPI and served by Uvicorn.
Metric	Uptime monitoring and error logs aiming for 99% up-time.
Priority	High
Rationale	Uvicorn provides real-time services with minimal downtime, though maintenance may occasionally cause delays.
Originator	Development Team

2.2.2 Module 2: Question Answering (QA) in User Mode

1. NFR2-001

Table 2.53: NFR2-001: Accuracy

Attribute	Description
Title	Accuracy
ID	NFR2-001
Version	1.0
Description	The system shall provide answers with at least 90% accuracy in text-based and visual question-answering using Hugging Face models and RAG.
Metric	Accuracy tests using a sample of 100 questions, measured against pre-defined answers.
Priority	High
Rationale	Accuracy of Hugging Face models may vary depending on training data, so 90% is a realistic threshold.
Originator	Development Team

2. NFR2-002

Table 2.54: NFR2-002: Scalability

Attribute	Description
Title	Scalability
ID	NFR2-002
Version	1.0
Description	The system shall handle up to 500 concurrent users without performance degradation, using LangChain and Vector Database for efficient retrieval operations.
Metric	Load testing to measure performance under 500 concurrent users.
Priority	Medium
Rationale	LangChain and Vector Database allow for efficient query handling but realistic constraints apply to vector search models.
Originator	Development Team

3. NFR2-003

Table 2.55: NFR2-003: Response Time

Attribute	Description
Title	Response Time
ID	NFR2-003
Version	1.0
Description	The system shall provide answers within 3-5 seconds for typical queries using Retrieval-Augmented Generation (RAG).
Metric	Response time logging to ensure average answers are provided within 3-5 seconds.
Priority	Medium
Rationale	FASTAPI pipelines for RAG introduce some overhead, so a 3-5 second response window is realistic.
Originator	Development Team

2.2.3 Module 3: Question Answering in Evaluation Mode

1. NFR3-001

Table 2.56: NFR3-001: Security

Attribute	Description
Title	Security
ID	NFR3-001
Version	1.0
Description	The system shall use SQLite for securely storing student data, including answers and evaluation results.
Metric	Security tests to ensure all student data is encrypted in transit and at rest.
Priority	High
Rationale	Database enhances security for sensitive student data while maintaining performance.
Originator	Development Team

2. NFR3-002

Table 2.57: NFR3-002: Timeliness

Attribute	Description
Title	Timeliness
ID	NFR3-002
Version	1.0
Description	The system shall provide real-time feedback for evaluation results, processed within 3-5 seconds using FASTAPI and LangChain for question generation.
Metric	Time-to-feedback will be monitored to ensure 95% of evaluations are processed within 5 seconds.
Priority	Medium
Rationale	Fast feedback is essential for an interactive learning experience, though question complexity may introduce slight delays.
Originator	Development Team

2.2.4 Module 4: Parental/Guardian Role in Evaluation Mode

1. NFR4-001

Table 2.58: NFR4-001: Accessibility

Attribute	Description
Title	Accessibility
ID	NFR4-001
Version	1.0
Description	The system shall comply with WCAG 2.1 standards for accessibility, ensuring full support for screen readers and easy-to-navigate interfaces in React.
Metric	Accessibility testing will ensure compliance with WCAG 2.1 standards.
Priority	High
Rationale	React supports responsive and accessible interfaces, enabling all users, including those with disabilities, to monitor student progress easily.
Originator	Development Team

2.2.5 Module 5: Audio Input and Output

1. NFR5-001

Table 2.59: NFR5-001: Compatibility

Attribute	Description
Title	Compatibility
ID	NFR5-001
Version	1.0
Description	The system shall support audio input/output using JavaScript and React across desktop, tablet, and smartphone devices.
Metric	Compatibility tests on 95% of supported devices to confirm smooth audio input/output.
Priority	Medium
Rationale	React and JavaScript allow for cross-platform compatibility, ensuring students can use the audio features seamlessly on any device.
Originator	Development Team

2. NFR5-002

Table 2.60: NFR5-002: Response Time

Attribute	Description
Title	Response Time
ID	NFR5-002
Version	1.0
Description	The system shall process audio input and provide audio output responses within 4-6 seconds using FASTAPI and Uvicorn.
Metric	Response time logging to track audio input/output, ensuring 95% of responses occur within 6 seconds.
Priority	Medium
Rationale	Audio queries may take slightly longer than text due to processing overhead, so a 4-6 second response time is realistic.
Originator	Development Team

2.2.6 Module 6: Dual Language Support

1. NFR6-001

Table 2.61: NFR6-001: Localization

Attribute	Description
Title	Localization
ID	NFR6-001
Version	1.0
Description	The system shall fully support both English and Urdu text, including right-to-left (RTL) support for Urdu using React.
Metric	Localization testing will ensure Urdu is rendered correctly, especially for RTL content, in 100% of cases.
Priority	High
Rationale	React has built-in support for RTL text rendering, ensuring an accurate representation of Urdu alongside English.
Originator	Development Team

2. NFR6-002

Table 2.62: NFR6-002: Language Processing Accuracy

Attribute	Description
Title	Language Processing Accuracy
ID	NFR6-002
Version	1.0
Description	The system shall ensure at least 85% accuracy for Urdu and English speech-to-text and text-to-speech features, using Hugging Face and LangChain models.
Metric	Accuracy tests using a sample size of 100 queries, with a target of 85% correct conversions for both languages.
Priority	Medium
Rationale	Due to the complexity of handling multiple languages, 85% accuracy is a realistic threshold for both speech-to-text and text-to-speech capabilities.
Originator	Development Team

2.2.7 Module 7: Database

1. NFR7-001

Table 2.63: NFR7-001: Scalability

Attribute	Description
Title	Scalability
ID	NFR7-001
Version	1.0
Description	The system shall scale to handle up to 100,000 student and parent records in SQLite, with minimal degradation in performance for retrieval queries.
Metric	Performance tests show the system remains functional with up to 100,000 records and less than 5% performance degradation.
Priority	High
Rationale	SQLite is designed for lightweight applications but can handle large datasets when properly optimized.
Originator	Development Team

2. NFR7-002

Table 2.64: NFR7-002: Data Integrity

Attribute	Description
Title	Data Integrity
ID	NFR7-002
Version	1.0
Description	The system shall ensure data integrity, with no data loss or corruption when storing and retrieving student profiles, progress, and evaluations using SQLite.
Metric	Integrity checks to ensure 100% of records are stored and retrieved without loss.
Priority	High
Rationale	SQLite provides reliable data handling, but additional validation mechanisms will ensure there is no data loss or corruption during transactions.
Originator	Development Team

2.3 Domain Model

A domain model is used to represent the core entities of a project or a system. Figure 2.2 highlights the key components such as Student, Parent/Guardian, Evaluation, Chatbot, and Language in our case. It helps us understand the interaction of all entities involved in SparkQuest.

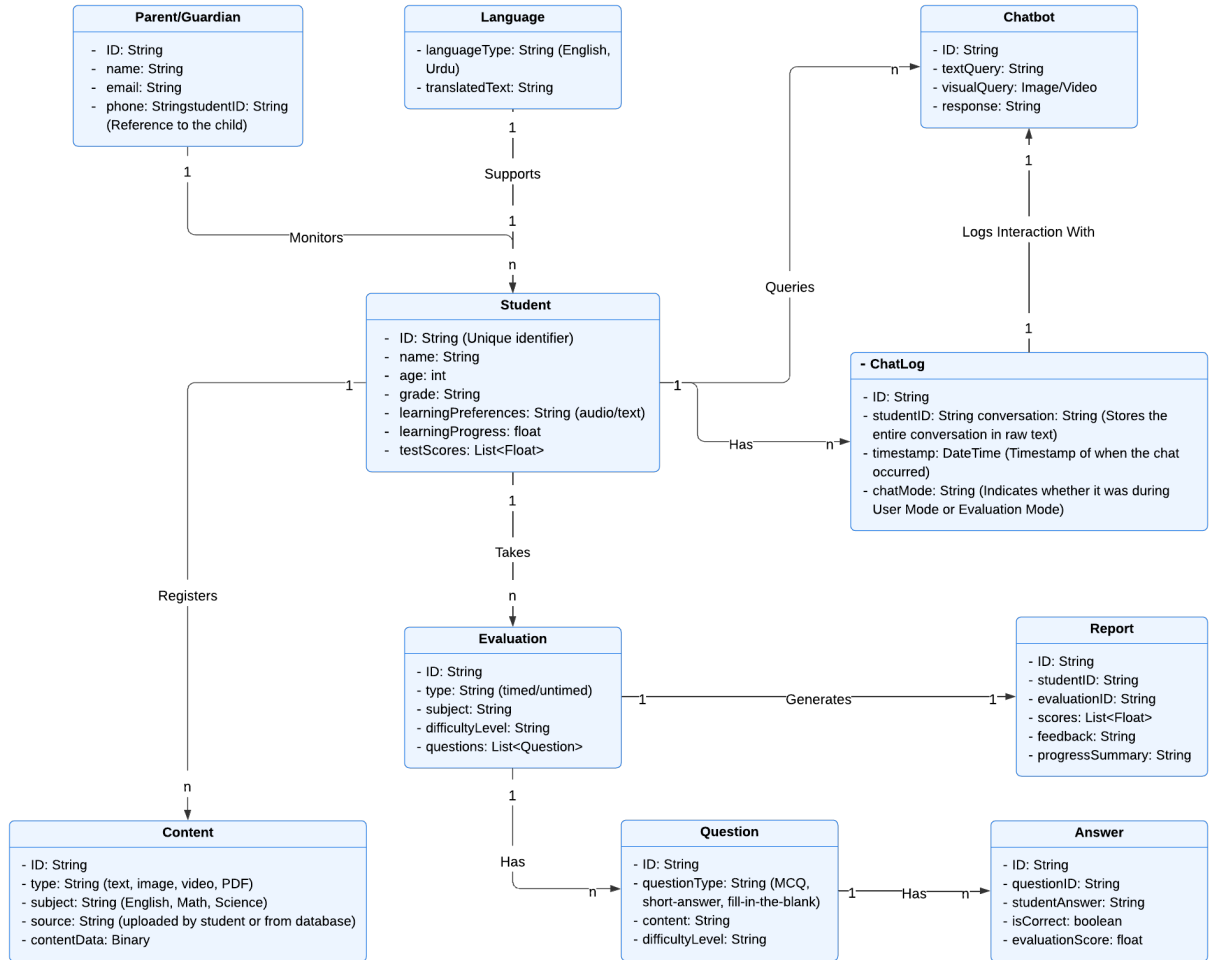


Figure 2.2: The Domain Model of SparkQuest

Chapter 3

System Overview

SparkQuest is an all-inclusive, AI-based educational tool meant for primary-grade learners. It makes use of RAG, speech-text, and text-speech technologies. There are also multiple modes including User Mode in which students can access educational content such as text, image, or video and ask through texts and even spoken questions which would be answered by AI. It also includes an Evaluation Mode where the student will be able to take quizzes and tests to evaluate his performance. The multilingual support of English and Urdu language along with the requirement of image and video as input to solve the problems become highly flexible in terms of learning.

Other than the features that will cater to the students, SparkQuest has developed a Parental Monitoring module where parents and guardians will be able to monitor their student's progress, access the chat history, and retrieve their evaluation results from a dashboard. Having a layered structure with presentation, business, and data layers, the system provides an ability to scale the system and make a proper management of data. It maintains the privacy of data records, quizzes, chats, and other contents while enabling asynchronous and synchronous conversations. The following chapter explains why SparkQuest is developed for young students with flexible learning preferences and parental control.

3.1 Architectural Design

The architecture of SparkQuest is based on a multi-tiered structure, which breaks the system into three distinct layers: the Presentation Tier, the Business Logic Tier, and the Data Tier. The Presentation Tier is the outermost and communicates between students, parents, and guardians through interfaces such as for progress and result evaluations. Here, at the Business Logical Tier, actual operations such as content management, question answering, evaluations and chat logging take place. Last but not least, the Data Tier is responsible for structurally storing information about students and the contents they are enrolled in,

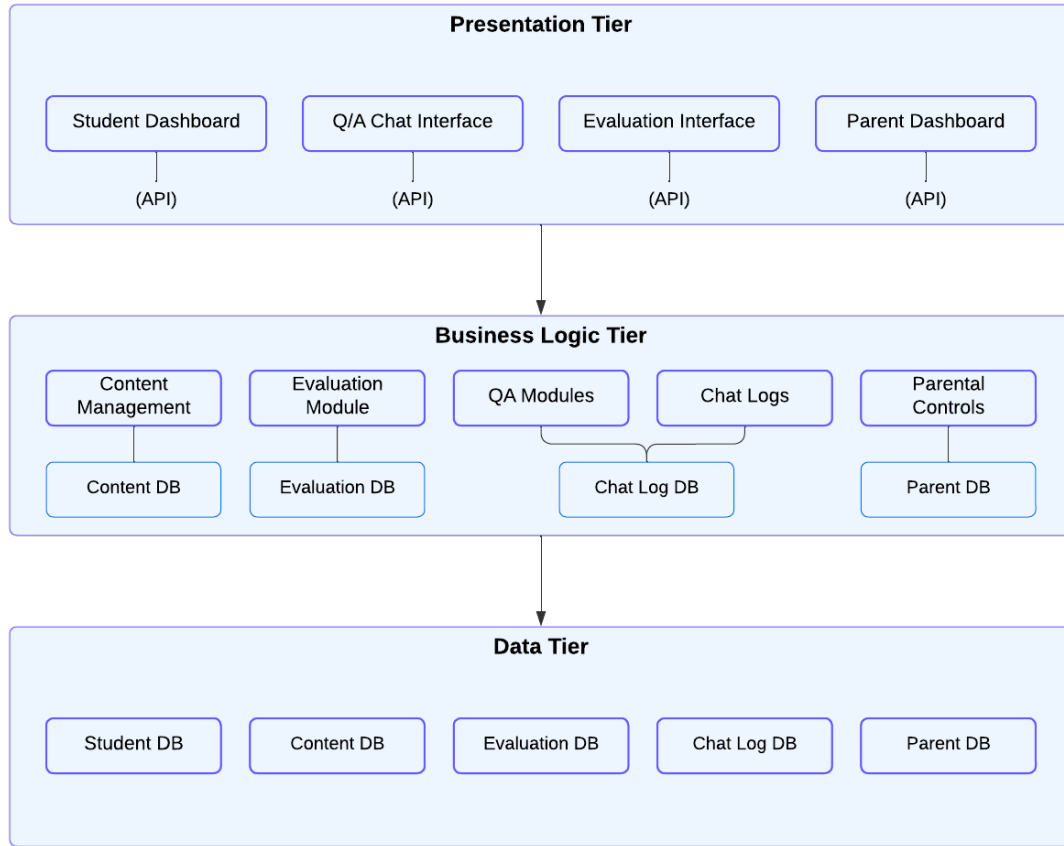


Figure 3.1: A Multi-Tiered Domain Model representation of SparkQuest

contents' evaluations, and content of chats initiated in the System. Such layers communicate with each other through APIs; thus, the overall architecture of the system is tiered, convenient to scale, and possesses smooth inter-tier communication. Figure 3.1 shows how these subsystems work in sync to ensure the smooth running of SparkQuest.

3.2 Design Models

3.2.1 Design Models for Procedural Approach

In the procedural approach, the design model focuses on accomplishing the task by using the defined steps, unlike the object-oriented design which focuses more on objects and their interactions. For SparkQuest, the procedural approach was better as it focuses more on the data flow and step-by-step execution of tasks divided into modular functions.

3.2.2 Activity Diagram

The activity diagram for SparkQuest in figure 3.2 illustrates the flow of three key processes: Student asking questions, student evaluation testing, and parental monitoring have been defined with a specific column for each of the involved entities. The system is divided into five layers: Student Mode, Guardian Mode, Presentation Tier, Business Logic Tier, and Data Tier. In each process, either the student or parent input is collected by the Presentation Tier and is then passed to the Business Logic Tier where it is processed by specific modules, the QA Module, the Evaluation Module, or the Parental Monitoring Module respectively. The data that is generated here is either saved or pulled from the Data Tier component, which handles essential stuff such as chat histories, evaluation outcomes, and reports.

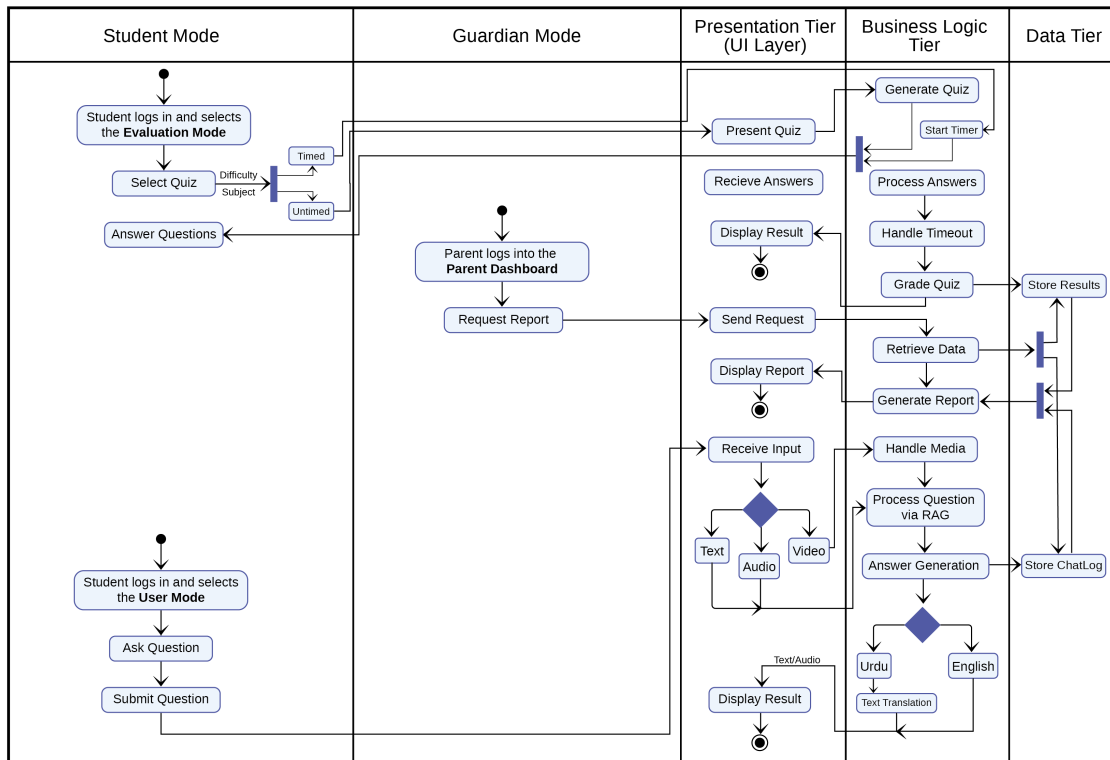


Figure 3.2: Activity Diagram of SparkQuest modeling the flow from one activity to another

3.2.3 Data Flow Diagram

The Data Flow Diagram for SparkQuest shows the flow of data across different modules of the system. It shows the interaction between users (students and parents/guardians) and

the system's core processes.

3.2.3.1 Level 0: SparkQuest as a Black Box

As shown in figure 3.3, Level 0 of SparkQuest is a black-box system shaped in a way that the users, such as students or parents provide a certain input and get a response in return. The students submit questions and Upload content and the parents request reports. The system gives students answers and evaluations from the uploaded query or content, whereas parents get reports and logs. At this stage, the system can be seen from the outside and the only factor contemplated is the exchange between the users of the service and the system. The unified data store is designed to store all chat logs evaluation content and progress data and make information available as required.

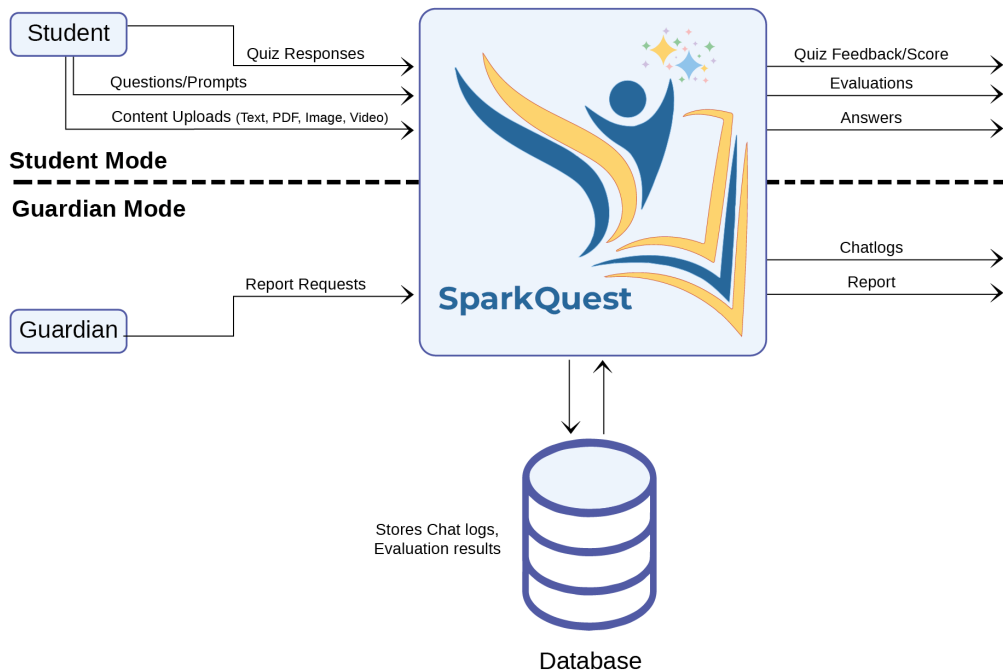


Figure 3.3: The Data Flow Diagram of SparkQuest at Level:0 where the system is a black box and the user inputs questions or documents and receives a relevant output.

3.2.3.2 Level 1: Modules Breakdown

At Level 1 the SparkQuest system has five critical components, Question Answering (QA), Evaluation, Content, Parental Monitoring, and Audio. These modules, as seen in figure 3.4, are designed to be functional units of the system that perform certain computations or operations. For instance, the QA module deals with the questions from students

while the Evaluation module deals with quizzes and grading as well as the Content Management module for uploaded content. All the above modules are connected to a single database where the data is stored and then passed on to other required modules.

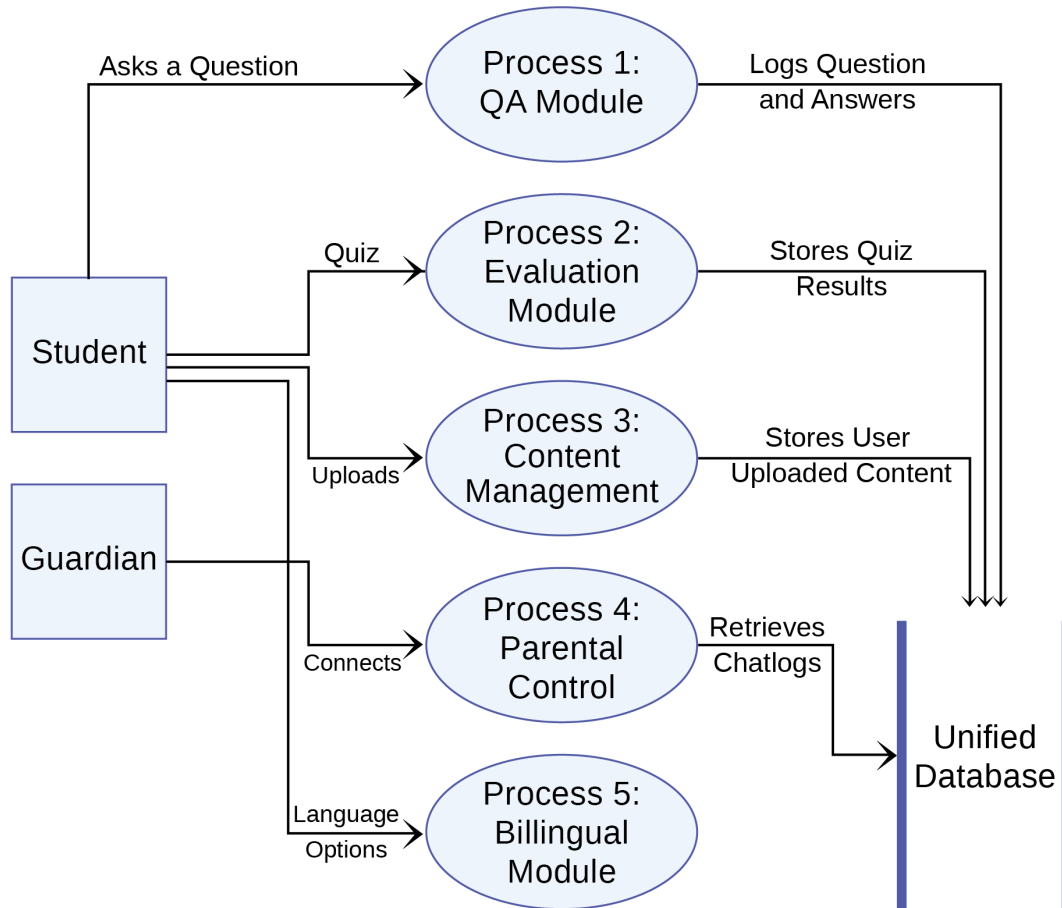


Figure 3.4: The Data Flow Diagram of SparkQuest at Level:1 The system is divided into five key modules and all are linked to a Database

3.2.3.3 Level 2: Sub-Processes within Modules

Level 2 of the Data Flow Diagram breaks down the general processes contained in each of the modules further into sub-processes as illustrated in figure 3.5. For example, in the QA module, sub-processes transform incoming elements from students, utilizing Retrieval Augmented Generation (RAG) on the question, and documenting the interaction in the Unified DB. Likewise in the Evaluation module of the system, the quizzes are administered, the responses from the students are also managed, the responses are then graded and the final results are stored and recorded for record purposes. Each of the sub-processes

is well connected to the Unified Database so that any data that is collected or required is well managed.

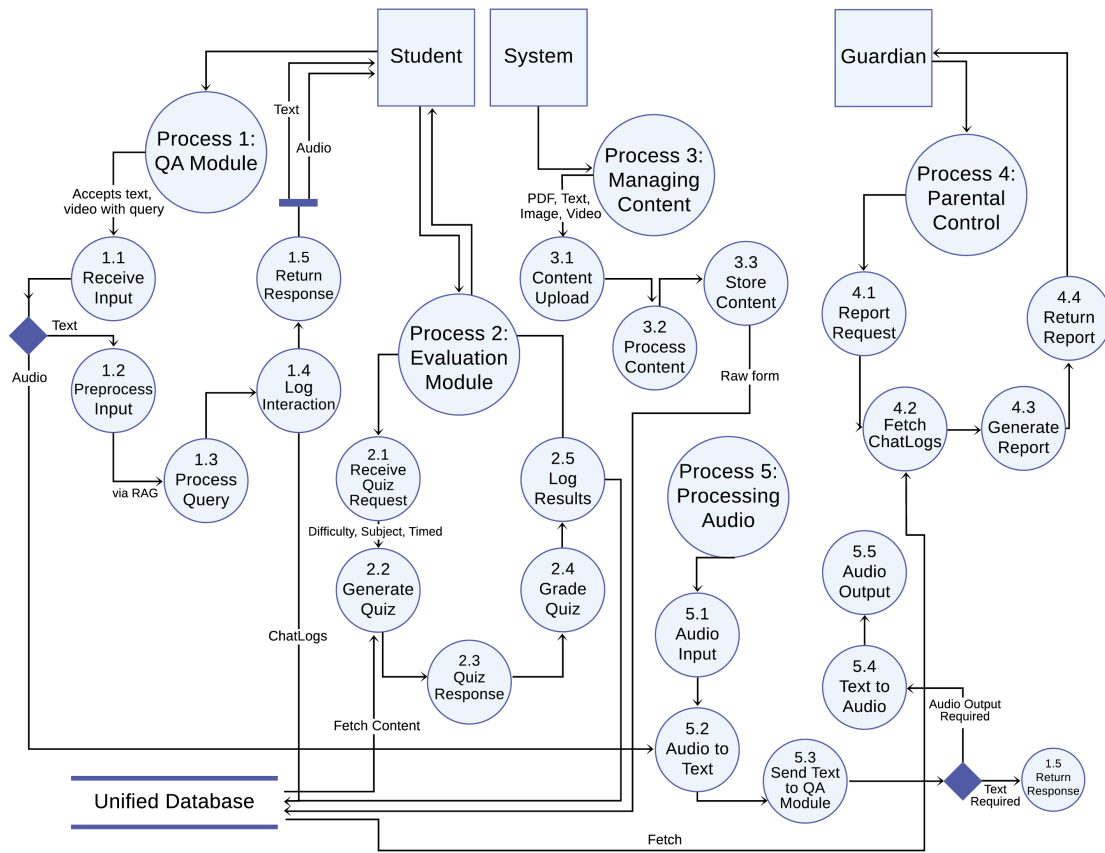


Figure 3.5: The Data Flow Diagram of SparkQuest at Level:2 where each module interacts with the Database to store or retrieve the data.

3.2.4 System-level Sequence Diagram

The System-Level Sequence Diagram for SparkQuest presents use cases that describe the type of communication between the main actors (students and parents), and core system parts including the QA Module, the Evaluation Module, the Content Management Module, the Parental Monitoring Module, the Audio Processing Module, and the Unified Database. This diagram shows how different operations take place within the system. Figure 3.6 shows how data flows and where the users will be interacting with the system at a general level. The processes of the system include questions posted by students while on the other hand, parents can request progress records from the various modules. It prescribes the flow of how each of the user requests flows through the layers of the system, from the presentation layer through the application business logic layer and then to the

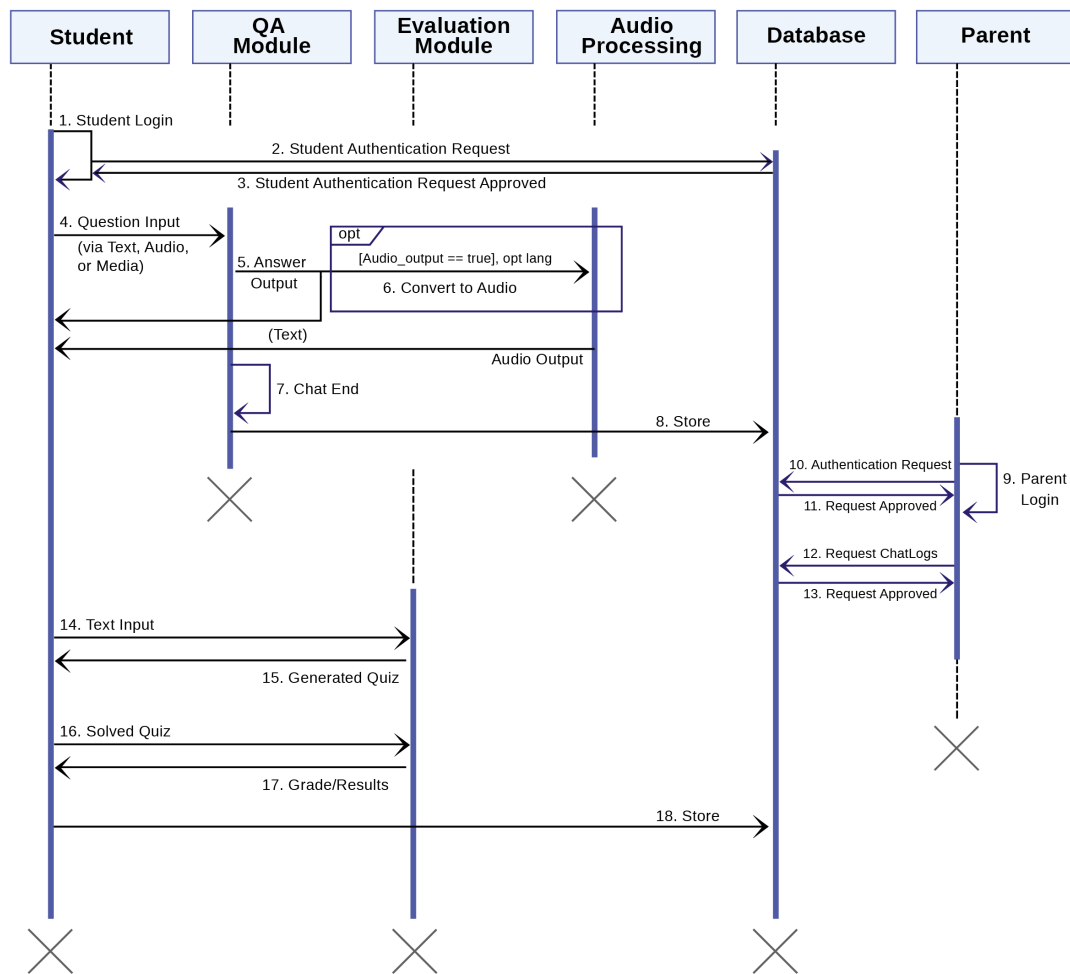


Figure 3.6: The System Level Sequence Diagram depicting the interactions between students, parents, and the main components (modules) of the system.

database management layer.

3.2.5 State Transition Diagram

The state transitions for SparkQuest depict how the system reacts to its various inputs as well as processes coming from students and parents as shown in figure 3.7. Beginning with the Idle state, the system is waiting. Whenever a student uploads content, asks a question, or initiates an evaluation, the system moves to other states such as Content Uploading, Question Processing, or Evaluation in Progress. When in the use of audio inputs, these are forwarded to the Audio Processing to avoid complications in conversion forms in compatibility with the media. Throughout the state transition, the Data Logging state is always recording the event, message, and answers or reports from the students and

guardians in the Database. The system then go back to the Idle state again in anticipation of the next input to be processed. All of these flows let different tasks be processed smoothly while keeping all the actions and responses in a single place.

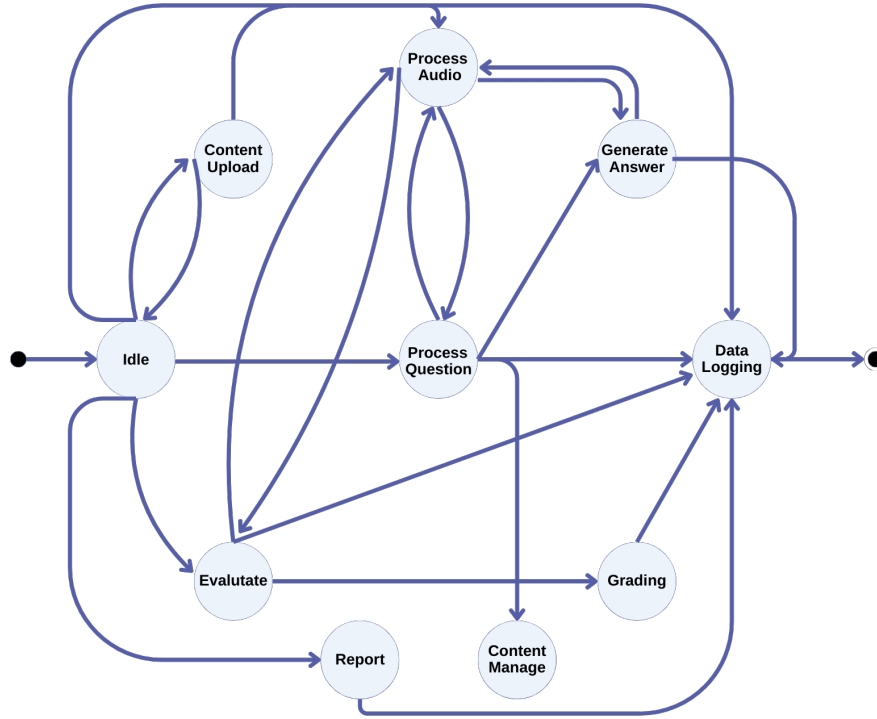


Figure 3.7: The Transitioning of Modules and Sub-processes within SparkQuest

3.3 Data Design

The database of SparkQuest consists of three main tables: Students, Guardians, and Chat-Log. The Students table keeps all the data related to each student separately using the fields email, password, grade, and a link to the guardian. The basic details of each guardian, including name, email, and password, are maintained in the Guardians table where a foreign key is used to link it to students with many-to-one relationships. The ChatLog table acts as a container for all various aspects of chat, including questions, answers, or even reports, as messages. The Entries in the ChatLog table are linked to the Students and the Guardians table respectively, with the help of foreign keys. This setup allows quick access to and easy chats data storage while keeping good relationships between students, guardians, and their chats, eliminating the need for different tables for various kinds of chats as in shown figure 3.8.

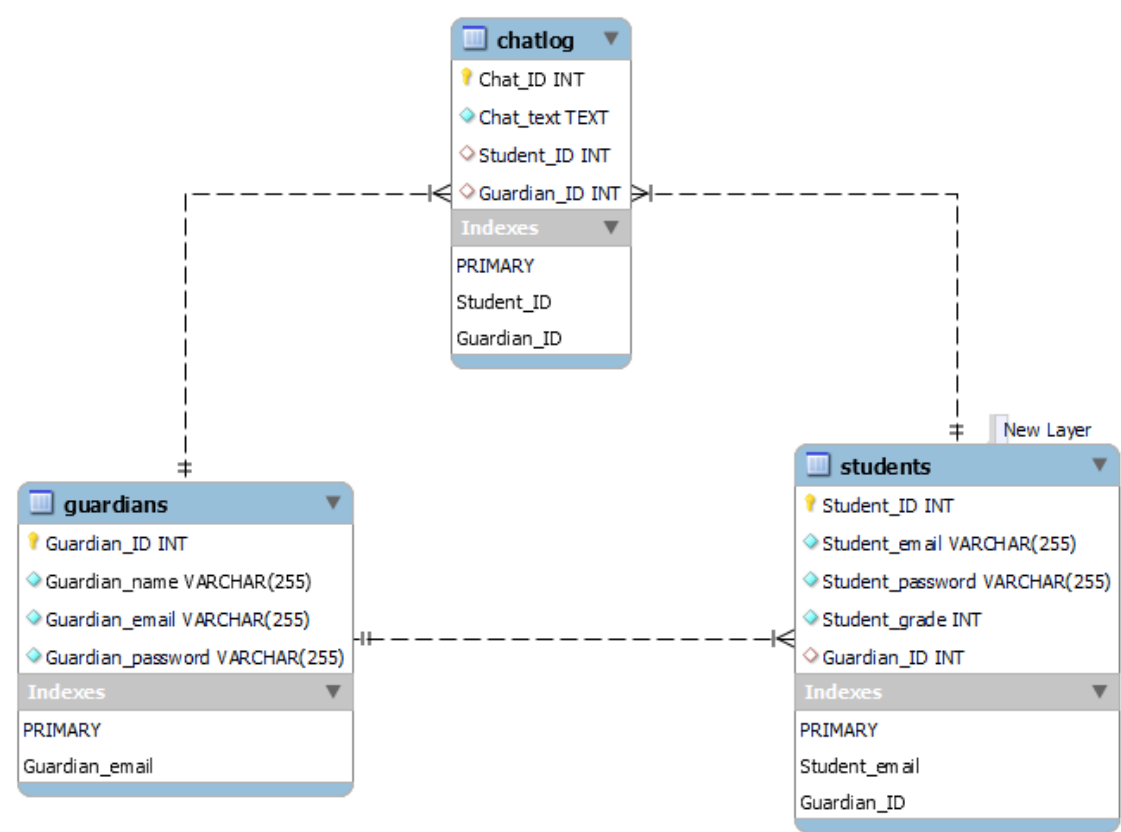


Figure 3.8: The Data Design of SparkQuest

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